

ENTERPRISE STANDARD OPERATING PROCEDURE SOP-4.2.1

Refinery Technical Services Directorate

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DOCUMENT CODE: SOP-4.2.1 (REV 4) | EFFECTIVE DATE: 01-APR-2025

STANDARD OPERATING PROCEDURE: SHELL & TUBE HEAT EXCHANGER INTEGRITY

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SECTION 4.2.1 - MANDATORY INSPECTION FREQUENCY:

For all Class-1 and Class-2 heat exchangers operating in corrosive sour crude service

(>1.5 wt% total sulfur), physical non-destructive inspection and ultrasonic tube bundle

evaluation shall be conducted at a strictly mandated interval not exceeding 12 months.

Extension beyond 12 months requires formal executive approval.

SECTION 6.1 - RETIREMENT THICKNESS CRITERIA:

Retirement thickness for carbon steel shell operating under sour crude service is 7.8 mm.

Equipment with remaining wall thickness below 8.5 mm must enter Enhanced Monitoring Zone.

SECTION 7.1 - TURNAROUND ADVANCE PLANNING:

Turnaround must be scheduled immediately when remaining equipment life falls below 18 months.